

Question

The crystal formed by three main group metal elements has the chemical formula X_2YZ ($M < 200$), where the mass fraction of X is 24.33%, and the mass fraction of Y is 12.86%.

In this crystal, Y and Z form a layered structure similar to graphite, with a ratio of $Y : Z = 1 : 1$ in each layer. The upper and lower layers cannot completely overlap and are arranged in an ABAB... sequence, where the Y atoms in the AB layers are fully aligned, but the Z atoms are entirely misaligned. X is located above and below the center of each hexagon.

The crystal can be divided into a hexagonal unit cell with Y at the vertices. The lattice parameters are $a = 504.9 \text{ pm}$, $c = 1010 \text{ pm}$, and the distance between the X atoms above and below each hexagonal ring is 343.8 pm .

According to the question, select the correct option from the following choices.

A. The components of the crystal contain elements from the third main group.

B. The components of the crystal contain elements from the fifth main group.

C. The components of the crystal contain elements from the sixth main group.

D. In the unit cell, the position $(\frac{1}{3}, \frac{2}{3}, 0.17)$ is occupied by a Mg atom.

E. In the unit cell, the position $(\frac{1}{3}, \frac{2}{3}, \frac{1}{2})$ is occupied by the In atom.

F. In the unit cell, the position $(\frac{2}{3}, \frac{1}{3}, 0)$ is occupied by the Sb atom.

G. The crystal density is 2.44 g/cm^3

H. The crystal density is $2.95\text{g}/\text{cm}^3$

I. The crystal density is $2.82\text{g}/\text{cm}^3$

Answer

Correct Answer: **I**

Detailed Explanation

The ratio of the relative atomic masses of the three elements is $X : Y : Z = (24.33\%/2) : 12.86\% : (1 - 24.33\% - 12.86\%) = 12.17 : 12.86 : 62.81$, indicating that Z is the heavier metal element. From $M < 200$, it is calculated that $Z < 125.62$.

Assuming Z is In , calculating $M = \frac{114.82}{62.81\%} = 182.8$, then $Y = 182.8 \times 12.86\% = 23.51$, $X = 182.8 \times 12.17\% = 22.25$, with no corresponding elements.

Assuming Z is Sn , calculating $M = \frac{118.71}{62.81\%} = 189.0$, then $Y = 189.0 \times 12.86\% = 24.31$, $X = 189.0 \times 12.17\% = 23.00$, corresponding to Mg, Na .

Assuming Z is Sb , calculating $M = \frac{121.76}{62.81\%} = 193.9$, then $Y = 193.9 \times 12.86\% = 24.93$, $X = 193.9 \times 12.17\% = 23.60$, with no corresponding elements.

Similarly, assuming Z is any other main-group metal element, no combination of elements satisfies the requirements. Thus, the crystal is Na_2MgSn .

CHECKPOINT

2 PTS

The crystal is Na_2MgSn

The vertices of the hexagonal unit cell are occupied by Mg atoms, with Sn atoms in the AB layers not overlapping. Therefore, one hexagonal unit cell contains one A layer and one B layer, and Na atoms are located between the layers, totaling two Na_2MgSn structural units. The fractional coordinates of the Mg atoms are $(0, 0, 0), (0, 0, \frac{1}{2})$.

$$\text{Unit cell mass } m = \frac{2 \times 189}{6.023 \times 10^{23}} = 6.28 \times 10^{-22} g.$$

CHECKPOINT

0.5 PTS

$$\text{Unit cell mass } m = 6.28 \times 10^{-22} g$$

$$\text{Unit cell volume } V = \frac{\sqrt{3}}{2} a^2 c = \frac{\sqrt{3}}{2} \times 504.9^2 \times 1010 = 2.23 \times 10^{-22} cm^3.$$

CHECKPOINT

0.5 PTS

$$\text{Unit cell volume } V = 2.23 \times 10^{-22} cm^3$$

$$\text{Crystal density } \rho = \frac{m}{V} = \frac{6.28 \times 10^{-22}}{2.23 \times 10^{-22}} = 2.82 g/cm^3.$$

CHECKPOINT

1 PTS

$$\text{Crystal density } \rho = 2.82 g/cm^3$$